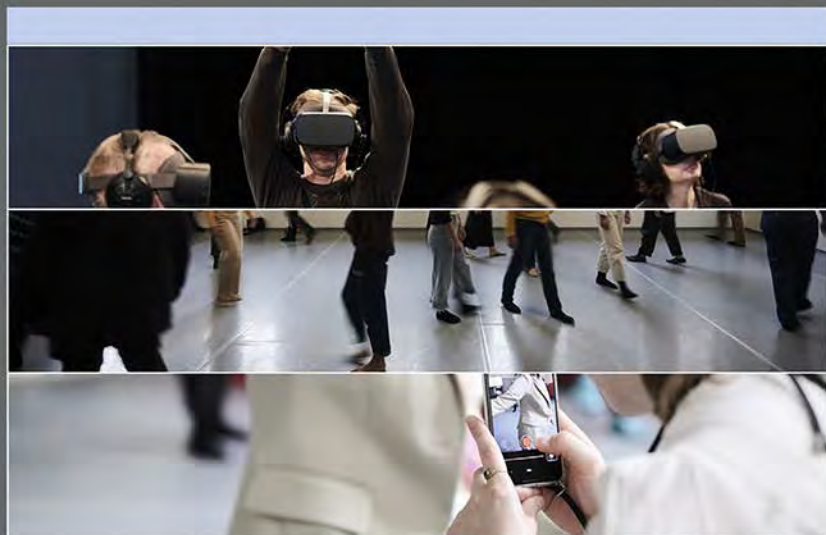




Yvonne Hardt, Marisa Joana Berg,
Anna Chwialkowska, Ulrike Nestler (Hg.)

Virtual Ecologies – Digitalitäten und Ökologien im Tanz

Yvonne Hardt, Marisa Joana Berg, Anna Chwialkowska, Ulrike Nestler (Hg.)
Virtual Ecologies – Digitalitäten und Ökologien im Tanz

Die Reihe wird herausgegeben von der Gesellschaft für Tanzforschung.

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Yvonne Hardt, Marisa Joana Berg, Anna Chwialkowska, Ulrike Nestler (Hg.)

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Virtu(re)al Bodies

Norbert Pape

In the following, I will argue that mixed reality¹ (MR) has as a distinctive feature the capacity to speak with and through the sensorimotor apparatus of the human body. Its goal is to invoke a sense of presence, space, and movement by addressing multiple senses simultaneously. MR has only very recently become widely accessible and its potential is yet hard to grasp. By enabling the physical manipulation of digital representations, it opens up, for example, new prospects for embodied learning and communication. As with other digital technologies, its capacity to predict, control, and modify human behavior presents an imminent danger, if used, for example, for political purposes or profit making. Furthermore, current applications stemming from the entertainment industry tend to support hegemonic forms of (dis)embodiment and bias. Nonetheless, and maybe all the more, it is crucial to develop and explore progressive and emancipatory uses of the technology, along with ethical considerations and guidelines. As a technology that interferes in the mind-body-world relation of its users, it provides a privileged site of experimentation with embodiment and thus also with what a body can do. I will conclude that knowledge of somatics, the body, and care stemming from dance are of great importance for the development of the field of MR, and that MR offers new fields of expression for dance makers, potentially reaching out to new audiences.

Neuroscience and branches of cognitive sciences and phenomenology that emphasize the role of the body and its actions in perception have played an important role in the development of virtual reality (VR). As Metzinger (2018) and Slater and Sanchez-Vives (2016) point out, a VR experience emerges out of

¹ Note that given the fact that MR has only been made available in the past few years, much of the literature I will refer to deals with virtual reality (VR). The distinctive feature of mixed reality is that the virtual objects are added to (usually layered onto) a depiction of the real surrounding environment.

the interplay of computer-generated audio-visual stimuli and the motor system of the body. It is in this sense that Oliver Grau inscribes VR into the art history of illusion and immersion by defining it as “a synthetic medium experienced polysensorily” (Grau 2003: 14), that attempts to weld traditional media together “to give the strongest impression possible of being at the location where the images are”. While this does differentiate the medium from audio-visual forms such as cinema and music, Thomas (2022) demonstrates that the path chosen by the VR industries is nonetheless ocular-centric. In most cases, the audiovisual stimuli are meant to override the physical body and environment, thereby potentially contributing to the reduction of the senses to the visual. Her research in experiential body-centered VR experiences, in which she explores the gap between seeing and feeling, as well as medical rehabilitation research prove that this does not have to be the case and open up exciting new fields of application for VR.

MR, which has only very recently become a standard capacity of most affordable head-mounted displays (HMDs), provides further means to address the body. It enables users to move freely and safely through ‘real’ spaces augmented with ‘virtual’ objects. Mixed reality can set bodies in motion and propose particular states of mind and qualities of space and presence, a capacity akin to dance. This is all the more interesting given that the novelty of the medium proves to attract audiences not necessarily inclined to attend dance workshops. If constructed with care, the experiences can be intimate, range from intuitive to challenging normative modes of perception, and have been proven to potentially have lasting effects on the users (cf. Madary/Metzinger 2016: 6).

1) Why is MR intrinsically kinesthetic?

Terms such as presence, body, space, sensory systems, perceptual systems, sensorimotor, embodiment abound in research papers about VR. Detailed descriptions of neuroscientific models of perception and cognition are given, especially relating to the question of how we gain a sense of space and of self, and the many ways in which the two can relate to each other.

The technical goal of VR is to immerse its users by substituting sensory input from the real world² with computer generated stimuli. It is worth noting here that while experimentation with other senses have been and are being performed, vision and sound are the ones most commonly addressed. Practical reasons might play into this, such as the costs of additional hardware (e.g. haptic gloves or suits), or the molecular nature of taste and smell versus the waveform of sound and light (cf. Lanier 2006). However, the restriction to the audio-visual is also evidently in line with norms of (dis)embodiment in our visually dominant culture. The following sections being dedicated to this subject matter, I want to first make the point that bodily movement is at the heart even of oculo-centric VR experiences. In order to create immersion, the computer-generated sensory input provided by the headset does not need to be a true reproduction of the full bandwidth stimuli provided by the real world: Only as much stimulation is needed as the mind requires to construct a sense of presence. Crucially, movement plays a role in this:

“In order to perceive we use our bodies in a natural way. We turn our head, move our eyes, bend down, look under, look over, look around, reach out, touch, push, pull, and doing all or some subset of these things simultaneously. Perception is a whole body action. Hence, the primary technological goal of VR is to realize perception through such natural sensorimotor contingencies to the best extent possible.” (Slater/Sanchez-Vives 2016: 5)

In short, the novelty of the medium is precisely that its effects are produced by the dialog between bodily movements and algorithms. The machine reads the movements of the body via sensors (e.g. sensors, cameras, microphone) and speaks to it via output devices (e.g. displays, speakers, haptic feedback). The digital objects, i.e. the ones perceptible only via the HMD, are thus not represented in the pictorial sense, but rather presented in a sculptural manner: They are meant to be apprehended through movement. Unlike 3D modeling on conventional computers, in which a person apprehends an object by rotating it on a screen, it is now the person itself that has to move around the object. It is worth noting here that the virtual object in itself is only stable on the level of code. Its appearance is modulated in real time with respect to the behavior of

2 Here and in the following, we will invoke the dichotomy of real and virtual for practical reasons, the real denoting that which is perceptible without HMDs, the virtual that which is perceptible only via the HMD.

the user and the intention of the artist. By no means do objects need to respect the laws governing the world, and in fact, the medium makes the thinkable experiential, and offers powerful modes of disrupting perception.

The intended effect of this body-machine-interaction is that it endows the virtual objects with an unusually high degree of presence. At the same time, one's own sense of presence and agency in the virtual world is heightened (compared, for example, with regular screens or projections). According to Metzinger, presence is a complex phenomenal quality, the three major dimensions of which are identification, self-location in a temporal frame of reference, and self-location in space. The interesting point here, he points out, is that human phenomenology can thus vary along a spectrum from realness to mind-dependence (cf. Metzinger 2018: 3). It is an experience co-produced by the body of the user and the technology, which can propose a vast range of levels of embodiment and "open the door to a deeper understanding of the mechanism of identification underlying the way in which a conscious subject of experience locates itself in time and space by identifying with a body" (Metzinger 2018: 7).

2) Body-centric approaches to VR

In answer to the question "Can we harness VR's potential to awaken new modes of perception that enable people to better know their body?" (Thomas/Glowacki 2018: 145), choreographer and filmmaker Dr. Lisa May Thomas reminds us of the obvious but easily overlooked fact that a head mounted display is first and foremost a blindfold (cf. Thomas 2018). As done in some forms of dance training, blindfolds can be used to de-emphasize visual form in favor of the kinaesthetic, proprioceptive and tactile senses which govern bodily position, movement, balance and touch. Medical research has indeed proven that VR can help train proprioception in some cases of motor control disorders, such as those resulting from a stroke, in which patients can be over-reliant on visual input (cf. Cho et al. 2014). In their rehabilitation exercises, Cho et al. "manipulate the visual feedback during upper-limb training and ask the subject to rely only on proprioception feedback" (Cho et al. 2014: 1). Similarly, Thomas and Glowacki inquired the gap between seeing and feeling by layering virtual environments onto the physical environment, seeking "a possibility for choices around ways in which the body can re-learn, through its plasticity, new modes of perception" (Thomas/Glowacki 2018: 20). They did so by designing a series of work-

shops in which the effect of discrepancies between that which was felt and that which was seen could be explored through movement.

3) Thoughts on working with mixed reality

Together with a team whose backgrounds span from dance, mathematics, sculpture, philosophy, spirituality to music³, I chose to approach the medium of MR transdisciplinarily with the aim to create a choreographic installation that would deepen our relation to our bodies and re-enchant the world. In the following, I will present tools we developed in the frame of the creation of the piece *Touching Clouds* (TC)⁴, contributing to the development of body-centric approaches to MR.

Touching Clouds (TC) is a choreographic mixed reality installation consisting of three to five headsets in a fairly large and fairly empty space. It invites the visitor to relate to and interact with a collection of animated virtual objects through movement and touch. One of the defining characteristics of the work is the range of reactions that objects have to touch.

Objects in VR are most often represented as meshes, i.e. surfaces defined by points connected by triangles. Interactions with these surfaces are most commonly reduced to actions such as bumping or grabbing (either with controllers or hands). In *Touching Clouds*, we opted for the computationally far more complex point cloud representation of objects, in which the surface is suggested by a multitude of points. Further, we developed software enabling tactile interaction with each individual point, using AI to track the hands of the user, along with the parallel processing capacity of graphic cards. This refined mode of interaction aimed at maximizing bodily immersion. We also experimented with animating the millions of points using movement principles derived from nature⁵ in order to alter the felt-sense of space.

3 Team: Simon Speiser, Norbert Pape, Franziska Aigner, Göksu Kunak, Hanako Hayakawa, Taís Lobo, Pêdra Costa, Austin Gross.

4 Premiered May 2023 in the frame of the *Moovy Dance Film Festival* and co-produced by *Theater im Depot Dortmund*. Documentation here: <https://norbertpape.github.io/digital.html> [online] [11.03.2024]

5 I hereby mean mathematical models developed to simulate or approximate natural phenomena such as the fluid dynamics, swarm behavior or the effect of wind on a field of grass. They have been widely used, and partly developed, for creating computer animations in films, anime and video games.

In order to maximize the engagement of the body and its articulation, research was done on many levels (music, sound design, narrative, character animation, dramaturgy), however, in the following, I will focus on the topics most specific to VR. As the work has been presented as an artistic project, so far, we have not yet prioritized a scientific and methodological analysis of its effects. The thoughts below are a synthesis of what we learned in making the work and insights we gained through numerous conversations with people that had experienced it.



Fig. 1: Testing Touching Clouds at Dock11, April 2023, (Simon Speiser)

Mixed reality and agency

Disconnected from their immediate surroundings, a person immersed in VR is very vulnerable. Protocols for ensuring the physical well-being of the users often involve isolating them and confining their movements to restricted spaces, but little is done to address the complexity of being exposed to the gaze of the others while immersed. In our experience, working with mixed reality proved

crucial for making the user feel safe to engage with the work in the presence of others. Factually, users became part of the work and perform long sequences of complex movements in front of others. They experienced little or no movement inhibition yet, interestingly, without forgetting the presence of the others. The visibility of the surrounding space in MR can stand in the way of some of VR's important ability to make *surreal* spaces experiential, such as the impressive infinite black void VR affords. In order to retain this capacity, we made the visibility of one's surroundings dependent on the height of the head relative to the floor. The closer one gets to the floor, the less of the real world one sees, but it reappears the moment one stands up. These types of correlations proved useful to give agency to the user: instead of being thrown into the void, they can choose to go there, gradually getting accustomed to their new spatial conditions.



Fig. 2: Tactile interaction with a point cloud in Touching Clouds, PAF 2023 (Norbert Pape)

Tactility and point clouds vs. meshes

Rarely used in standalone VR apps due to the required computing power, we opted for point cloud representation of objects. We had to heavily optimize and extend Keijiro Takahashi's point cloud renderer (cf. Takahashi 2021), an open-source Unity asset. For example, we generated GPU based algorithms (so called

shaders) that enable high-definition tactile modes of interaction with the individual points of the point cloud, fingers leaving traces on objects touched (cf. Pape 2023). Further, we endowed the points with the capacity to fall like sand, or float up like dust, or slowly move back to their original position, giving an impression of a wound healing. Besides reportedly inducing synaesthetic responses in the users (heat or tingling sensation in the fingers), they also evoked strong emotional responses. We suspect this is due to the particular materiality suggested by point clouds, which resemble weightless and ephemeral things we very well have interacted with physically before, such as smoke, dust, fog, spiderwebs, and similar.



Fig. 3: The virtual object missing in the reflection in the mirror, scene from Touching Clouds, PAF 2023 (Norbert Pape)

Defying laws of nature and space

With floating stones, some bouncing off the walls, others flying through them, objects inside or behind walls, we created a tension between the real and the virtual environments, inviting the user to exist in different spaces at the same

time. This proved to entice an exploration not only of each of the multiple spaces, but also of their relation, often resulting in people touching walls, seemingly looking through the floor. Further, the fact virtual objects would respond only to head and hands but fly through all other parts of the body led to explorations of what one's augmented body can do: People touching themselves, trying to manipulate invisible objects with various parts of their bodies. Conversations revealed that there was a pleasure in experiencing incongruities between layered spaces, such as stones flying through cement walls, or points falling like dust through the hard floor. We suspect that these incongruities helped induce a dream-like state in which one lost a sense of time, and that some were busy trying to understand, in some ways to reverse-engineer, the virtual world and the technology behind it.

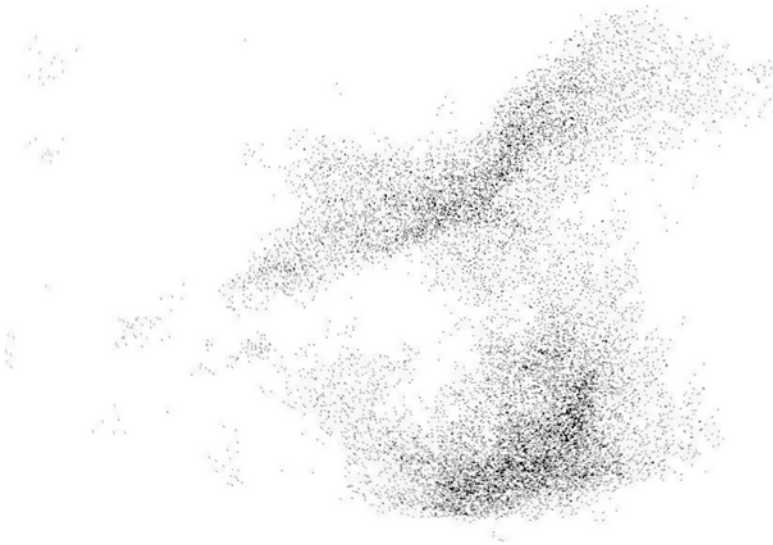


Fig. 4: Boids algorithm applied to a point cloud, turning it into a cloud-like phenomenon, scene from Touching Clouds, PAF 2023 (Norbert Pape)

Mimicking organic processes

Boids, an algorithm for flock simulation, and Perlin noise, a continuous function generating nonrepeating organic looking patterns, were developed for the sake of increasing the organicity and immersive qualities of video games and CGIs, and also proved useful in TC. Similar to the point clouds, subtle movements and kinetic phenomena familiar from the real world (e.g. flocks of birds, clouds of dust, leaves in the wind) proved to increase immersion. Further exploring the use of simulation of natural kinetic processes to enhance the machine/human interface has become the starting of a new creation process of ours.

4) Conclusion

“[...] by changing space, by leaving the space of one’s usual sensibilities, one enters into communication with a space that is psychically innovating [...] For we do not change place, we change our nature.” (Bachelard, 1994: 206)

Since the early 1990s, artists have been experimenting with machine/human interfaces that utilize intuitive physical processes, such that “the observer’s unconscious connects to the virtual space in a much more intense way than with a joystick or a mouse. [...] This physically intimate design of the human-machine interface gives rise to such immersive experiences that the artist speaks of reaffirming the participants’ corporeality.” (Grau 2003: 198) Designing such interfaces and experiences requires not only tech-related skills. Whether the aim is to challenge or entertain the users, the design process requires profound knowledge of what makes us feel safe, how we direct our attention and make sense of things, how to make and shape spaces and how to invite others into them.

Without losing sight of the dangers entailed by vanishing interfaces, it is of utmost importance that we learn both how machines creep under our skin and how to make use of them in ways beneficial to sustaining life on earth. We have outlined three approaches, the first invested in gaining a more fine-grained understanding of processes of embodiment in order to extend our modes of being in the world (cf. Metzinger 2018: 6), the second developing body-centric approaches to VR by activating bodily perception in the gaps between layered

realities (cf. Thomas 2022), and the third, ours, by using mixed reality to ease the embodied way into other-worldly emotionally textured spaces. MR opens new prospects for embodied learning, as learners can now manipulate digital representations of abstract concepts and make sense of them through sensorimotor stimulation (cf. Chatain 2023). Research on avatar embodiment provides evidence that behavior while in the virtual environment can have a lasting psychological impact after subjects return to the physical world, and thus influence behavioral responses (cf. Slater/Sanchez-Vives 2016). Where else can these walks into abstraction take us? How do we want to dance with the machines?

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